

Ergodicity is complicated, let us only focus on the ergodicity of a wide stationary process.

The define of a wide stationary process is:

$$\begin{cases} E_w(X(w, t)) = Const \\ Cov_w(X(w, t_1), X(w, t_2)) = f(t_2 - t_1) \\ E_w(X^2(w, t)) < +\infty \end{cases}$$

So, we can simply define ergodicity as:

$$E_w(X(w, t)) = E_t(X(w, t))$$

It seems that *L. H. S.* will leave t and *R. H. S.* will leave w , but if we take a closer look, it is not that case.

Wide stationary process tells us that:

$$L. H. S. = Const$$

Ergodicity means that $E_t(X(w, t))$ must be un-related with w and be the same as $E_w(X(w, t))$.

Further reading: section 4.2 of 《随机过程》, 方兆本。